TP2/3: Représentation graphique de données génomiques – Utilisation de Circos

(2026)

Objectifs :

-Apprendre à manipuler des données génomiques (et Circos) pour représenter:

- des données d'annotation de génomes
- des données structurales (ex: régions dupliquées)
- …

- **3 génomes à analyser (1/binôme)**

Arabidopsis thaliana
(Brassicaceae)

2n=2x=10

Oryza sativa
(Poaceae)

2n=2x=24

Sorghum bicolor
(Poaceae)

2n=2x=20

- **Analyses sur Genouest (id/pwd individuels)**

- **CC2 : Rapport de 2 pages + tous les scripts**
  (archive .tar.gz à déposer sur Moodle)

- **Données** /home/genouest/mob/asalmon/TP_EDG/data

- **1 dossier / espèce**

Athaliana_167_TAIR10.fa
TAIR10_GFF3_genes_transposons.gff.txt
At_At.aligncoords.gcoords.txt
Athaliana_167_TAIR10.annotation_info.txt
TAIR10_Transposable_Elements.txt

- Les séquences (*.fa)
- Les annotations structurales (*.gff*)
- Les annotations fonctionnelles (*.annotation_info.txt)
- Données de synténie intra-génomique (*.gcoords.txt)

Osativa_323_v7.0.hardmasked.fa
Osativa_323_v7.0.gene_exons.gff3
Os_Os.aligncoords.gcoords.txt
Osativa_323_v7.0.annotation_info.txt

Sbicolor_313_v3.0.hardmasked.fa
Sbicolor_313_v3.1.gene_exons.gff3
Sb_Sb.aligncoords.gcoords.txt
Sbicolor_313_v3.1.annotation_info.txt

- **Objectif: Représenter graphiquement (avec Circos):**

1- Les chromosomes

2- La densité en éléments transposables

3- La densité en gènes

4- Données Bonus (libre)

5- La synténie intragénomique

- **Objectif: Représenter graphiquement (avec Circos):**

Données brutes	Script#	Fichier formaté pour circos pour représenter:
*.fasta	Script 1	1- Les chromosomes
*.gff3 (At) *.hardmasked.fa (Os-Sb)	Script 2	2- La densité en éléments transposables
*.gff3	Script 3	3- La densité en gènes
*.**	Script 4	4- Données Libres
*. aligncoords.gcoords.txt	Script 5	5- La synténie intragénomique

Circos

https://circos.ca

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CANADA'S MICHAEL SMITH GENOME SCIENCES CENTRE

Martin Krzywinski science + art

BC CANCER RESEARCH INSTITUTE
Provincial Health Services Authority

CIRCOS.

GUIDE IMAGES SOFTWARE DOCUMENTATION PRESENTATIONS NEWS CITATIONS SUPPORT

DATA VISUALIZATION CIRCULAR APPROACH FEATURES GENOMIC DATA GENERAL DATA TABULAR VISUALIZATION PUBLISHED IMAGES

Circos > Introduction

WHAT IS CIRCOS?

CIRCULAR VISUALIZATION

Circos is a software package for visualizing data and information. It visualizes data in a circular layout — this makes Circos ideal for exploring relationships between objects or positions. There are other reasons why a circular layout is advantageous, not the least being the fact that it is attractive.

Circos is ideal for creating publication-quality infographics and illustrations with a high data-to-ink ratio, richly layered data and pleasant symmetries. You have fine control each element in the figure to tailor its focus points and detail to your audience.

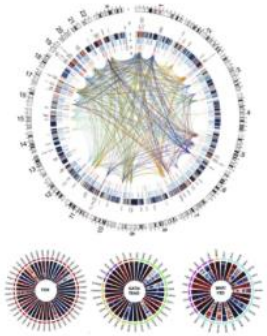
Circos https://circos.ca

The content on this page is old.

CIRCOS IMAGES IN SCIENTIFIC LITERATURE

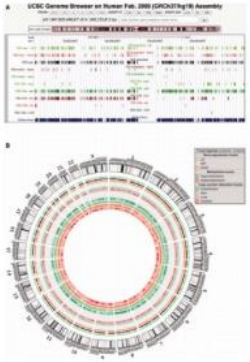
For a list of papers that use Circos, see Circos citations.

The images here and the list of citations hasn't been updated for quite some time. But if you're looking for examples of how Circos can be used, there's something here for everyone.

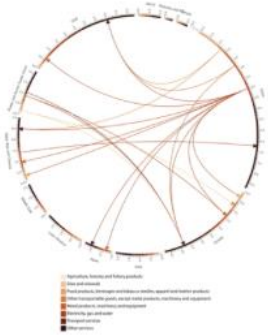


▲ **1** · 1 Dec 2013 | Saben J, Zhong Y, McKelvey S *et al.* (2014) A comprehensive analysis of the human placenta transcriptome *Placenta* **35**:125-131.

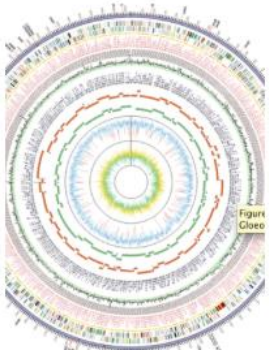
▲ **2** · 25 Oct 2013 | Bollongino R, Nehlich O, Richards MP *et al.* (2013) 2000 years of parallel societies in Stone Age Central Europe *Science* **342**:479-481.

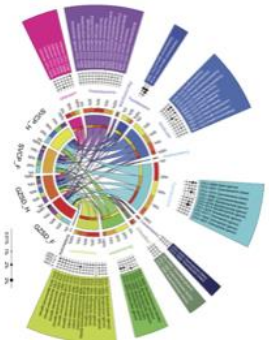


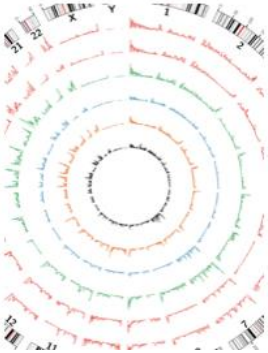
▲ **3** · 25 Oct 2013 | Dayem Ullah AZ, Cutts RJ, Ghetia M *et al.* (2013) The pancreatic expression database: recent extensions and updates *Nucleic Acids Res*

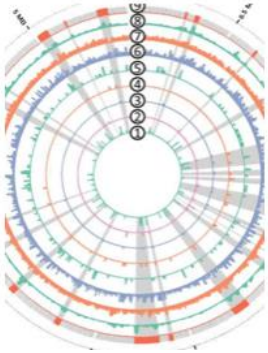


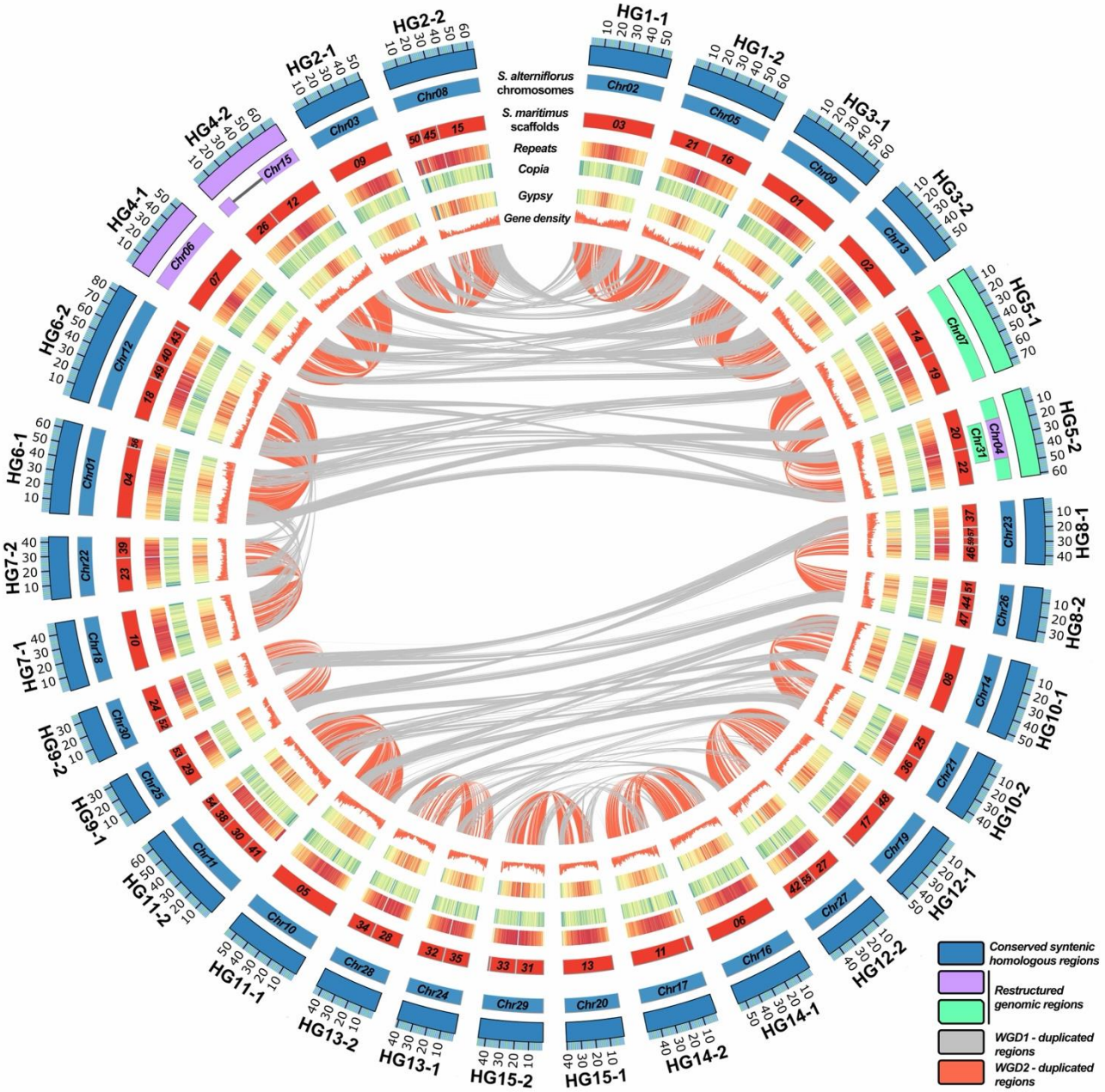
▲ **4** · 23 Oct 2013 | Kanemoto K, Moran D, Lenzen M *et al.* (2013) International trade undermines national emission reduction targets: New evidence from air pollution *Global Environmental Change*










S. alterniflorus chromosomes
S. maritimus scaffolds
Repeats
Copia
Gypsy
Gene density
HG1-1
HG1-2
HG2-1
HG2-2
HG3-1
HG3-2
HG4-1
HG4-2
HG5-1
HG5-2
HG6-1
HG6-2
HG7-1
HG7-2
HG8-1
HG8-2
HG9-1
HG9-2
HG10-1
HG10-2
HG11-1
HG11-2
HG12-1
HG12-2
HG13-1
HG13-2
HG14-1
HG14-2
HG15-1
HG15-2
Conserved syntenic homologous regions
Restructured genomic regions
WGD1 - duplicated regions
WGD2 - duplicated regions

- **Scripts circos exemple dans** /asalmon/TP_EDG/ scripts

circos_bash.sh → Lanceur (**sbatch circos_bash.sh**)

karyotype.txt
plot_1.txt
plot_2.txt
links.txt
→ Données formattées pour circos

circos.conf → **Fichier « maître » de configuration**

ideogram.conf
ideogram.label.conf
ideogram.position.conf
ticks.conf
bands.conf
→ Fichiers de configuration

circos.png
circos.svg
→ Fichiers de sortie

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5- La synténie intragénomique

1- Les chromosomes

Éditer le fichier **karyotype.txt**

chr	-	Chr1	Chr1	1	15000000	blues-6-seq-5
chr	-	Chr2	Chr2	1	15000000	blues-6-seq-5
chr	-	Chr3	Chr3	1	15000000	blues-6-seq-5
chr	-	Chr4	Chr4	1	15000000	blues-6-seq-5
chr	-	Chr5	Chr5	1	15000000	102,255,178

Début

fin

couleur

ID des chromosomes

1- Les chromosomes

Éditer le fichier karyotype.txt

chr	-	Chr1	Chr1	1	15000000	blues-6-seq-5
chr	-	Chr2	Chr2	1	15000000	blues-6-seq-5
chr	-	Chr3	Chr3	1	15000000	blues-6-seq-5
chr	-	Chr4	Chr4	1	15000000	blues-6-seq-5
chr	-	Chr5	Chr5	1	15000000	102,255,178

…à l'aide du fichier en fasta contenant les séquences génomiques

```
>Chr1 CHROMOSOME dumped from ADB: Feb/3/09 16:9; last updated: 2007-12-20
CCCTAAACCCTAAACCCTAAACCCTAAACCTCTGAATCCTTAATCCCTAAATCCCTAAATCTTTAAATCCTACATCCAT
GAATCCCTAAATACCTAATTCCCTAAACCCGAAACCGGTTTCTCTGGTTGAAAATCATTGTGTATATAATGATAATTTT
ATCGTTTTTATGTAATTGCTTATTGTTGTGTGTAGATTTTTTTAAAAATATCATTTGAGGTCAATACAAATCCTATTTCT
TGTGGTTTTCTTTCCTTCACTTAGCTATGGATGGTTTATCTTCATTTGTTATATTGGATACAAGCTTTGCTACGATCTA
CATTTGGGAATGTGAGTCTCTTATTGTAACCTTAGGGTTGGTTTATCTCAAGAATCTTATTAATTGTTTGGACTGTTTA
TGTTTGGACATTTATTGTCATTCTTACTCCTTTGTGGAAATGTTTGTTCTATCAATTTATCTTTTGTGGGAAAATTATT
TAGTTGTAGGGATGAAGTCTTTCTTCGTTGTTGTTACGCTTGTCATCTCATCTCTCAATGATATGGGATGGTCCTTTAG
CATTTATTCTGAAGTTCTTCTGCTTGATGATTTTATCCTTAGCCAAAAGGATTGGTGGTTTGAAGACACATCATATCAA
AAAAGCTATCGCCTCGACGATGCTCTATTTCTATCCTTGTAGCACACATTTTGGCACTCAAAAAAGTATTTTTAGATGT
TTGTTTTGCTTCTTTGAAGTAGTTTCTCTTTGCAAAATTCCTCTTTTTTTAGAGTGATTTGGATGATTCAAGACTTCTC
GGTACTGCAAAGTTCTTCCGCCTGATTAATTATCCATTTTACCTTTGTCGTAGATATTAGGTAATCTGTAAGTCAACTC
ATATACAACTCATAATTTAAAATAAAATTATGATCGACACACGTTTACACATAAAATCTGTAAATCAACTCATATACCC
GTTATTCCCACAATCATATGCTTTCTAAAAGCAAAAGTATATGTCAACAATTGGTTATAAATTATTAGAAGTTTTCCAC
TTATGACTTAAGAACTTGTGAAGCAGAAAGTGGCAACACCCCCCACCTCCCCCCCCCCCCCCCCACCCCCCAAATTGAGA
AGTCAATTTTATATAATTTAATCAAATAAATAAGTTTATGGTTAAGAGTTTTTTACTCTCTTTATTTTTCTTTTTCTTTT
TTGAGACATACTGAAAAAAGTTGTAATTATTAATGATAGTTCTGTGATTCCTCCATGAATCACATCTGCTTGATTTTTC
TTTCATAAATTTATAAGTAATACATTCTTATAAAATGGTCAGAGAAACACCAAAGATCCCGAGATTTCTTCTCACTTAC
TTTTTTTCTATCTATCTAGATTATATAAATGAGATGTTGAATTAGAGGAACCTTTGATTCAATGATCATAGAAAAATTA
GGTAAAGAGTCAGTGTCGTTATGTTATGGAAGATGTGAATGAAGTTTGACTTCTCATTGTATATGAGTAAAATCTTTTC
TTACAAGGGAAGTCCCCAATTGGTCAACATGTGAAAGCACGTGTCATGTTCTTACTTTTGTTTGGGTAATCTTCTAATT
ACTGTATATGGAAGATGTGAATGAAGTTTTGGTCCTGAATGTGGCCAAGGTTCCGTCATTTGGAGATACGAAATCAAAT
CTCCTTTAAGATTTTGTTTTTATAATGTGTTCTTCCATCCACATCTATCTCCATATGATATGGACCATATCATACATCA
TCATTTGTCCAAATGCATGAATGAATTTGGAAATAGGTACGAGAATGCCAACAATGACAAGAAGGGATCAAAGACAGTT
TTTAAAACAATATTTTACAGGGTTTTAATCTAATTCTAAGTTTTGGTCACTCACTTTGTTAAAAGAATAATTCAGTGTC
TGGACACTAAAATCTTCCAAAAACCCCATATACATATATGCTATTTCGATACTTATATTTATTTACTCAGCATAAAAAA
TATTAACCATGTATTCATAGTAAAATGTTTCATGTGATATCAAACCAGCGACAACAAAAGTATTATTCCCCTCATTATG
TTTGACTCCTATTATATTTTTATTTTAATTTTTTTCACTATCATCTTTCTTGCAATGAAAGTCCCATATATTGGTCAAC
ATTTCAAACCACTTGTTCTCTTTTATGTTTTGGTAAGAGCTATCTTCTAAATTTATAATACGCATAAATTCAAAAGTAA
AAGAAAATTTTGGTCATGAATGTTGTTTAAGTCATTTGGAGATACGAAATCAAATCTCCTTGTAGATTTTGTTTTTAGA
```

Attention! ne représenter que les chromosomes (pas les contigs)

- **Objectif: Représenter graphiquement (avec Circos):**

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5- La synténie intragénomique

2- La densité en éléments transposables

Éditer un fichier texte tabulé <TE_density>.txt **Par fenêtre de 100kb**

Chr1	1	100000	10832
Chr1	100001	200000	1683
Chr1	200001	300000	2231
Chr1	300001	400000	222
Chr1	400001	500000	3697
Chr1	500001	600000	5195
Chr1	600001	700000	6833
Chr1	700001	800000	456
Chr1	800001	900000	1701
Chr1	900001	1000000	11083
Chr1	1000001	1100000	0
Chr1	1100001	1200000	0
Chr1	1200001	1300000	9125
Chr1	1300001	1400000	2304
Chr1	1400001	1500000	8332
Chr1	1500001	1600000	3713
Chr1	1600001	1700000	1451

ID des chromosomes

Début

fin

Valeur de densité

2- La densité en éléments transposables

Éditer un fichier texte tabulé <TE_density>.txt **Par fenêtre de 100kb**

…soit à l'aide du fichier en fasta contenant les séquences génomiques « hardmasked »

```
>Chr1
NNNN
```

Le format GFF3

Liste et caractéristiques des "objects" dans un génome

General Feature Format 3

Features = "élément"
3 niveaux hiérarchiques :
gènes - **transcripts** - **exons**

```
##gff-version 3
ctg123 example gene            1050 9000 . + . ID=EDEN;Name=EDEN;Note=protein kinase
ctg123 example mRNA            1050 9000 . + . ID=EDEN.1;Parent=EDEN;Name=EDEN.1;Index=1
ctg123 example five_prime_UTR  1050 1200 . + . Parent=EDEN.1
ctg123 example CDS             1201 1500 . + 0 Parent=EDEN.1
ctg123 example CDS             3000 3902 . + 0 Parent=EDEN.1
ctg123 example CDS             5000 5500 . + 0 Parent=EDEN.1
ctg123 example CDS             7000 7608 . + 0 Parent=EDEN.1
ctg123 example three_prime_UTR 7609 9000 . + . Parent=EDEN.1
```

1	2	3	4	5	6	7	8	9
Séquence	**Source**	**Elément**	**Start**	**End**	**Score**	**Brin**	**Phase**	**Attributs**
Nom de la séquence	Nom de la méthode, de l'outil	Type d'élément	Position de l'élément le long de la Séquence [1]		Score de l'élément selon la Source [2]	Sens de l'élément (forward ou reverse)	Cadre de lecture (ORF)	Autres caractéritiques **ID** **Name** **...**

- **Objectif: Représenter graphiquement (avec Circos):**

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3- La densité en gènes

Éditer un fichier texte tabulé <gene_density>.txt

Par fenêtres (non chevauchantes) de 100kb

Chr1	1	100000	10832
Chr1	100001	200000	1683
Chr1	200001	300000	2231
Chr1	300001	400000	222
Chr1	400001	500000	3697
Chr1	500001	600000	5195
Chr1	600001	700000	6833
Chr1	700001	800000	456
Chr1	800001	900000	1701
Chr1	900001	1000000	11083
Chr1	1000001	1100000	0
Chr1	1100001	1200000	0
Chr1	1200001	1300000	9125
Chr1	1300001	1400000	2304
Chr1	1400001	1500000	8332
Chr1	1500001	1600000	3713
Chr1	1600001	1700000	1451

ID des chromosomes

Début

fin

Valeur de densité

3- La densité en gènes

Éditer un fichier texte tabulé <gene_density>.txt **Par fenêtre de 100kb**

…à l'aide du fichier .gff contenant la position des gènes

```
##gff-version 3
##annot-version v7.0
##species Oryza sativa
Chr1	phytozomev11	gene	2903	10817	.	+	.	ID=LOC_Os01g01010.MSUv7.0;Name=LOC_Os01g01010
Chr1	phytozomev11	mRNA	2903	10817	.	+	.	
ID=LOC_Os01g01010.1.MSUv7.0;Name=LOC_Os01g01010.1;pacid=33123311;longest=1;Parent=LOC_Os01g01010.MSUv7.0
Chr1	phytozomev11	exon	2903	3268	.	+	.	ID=LOC_Os01g01010.1.MSUv7.0.exon.1;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	five_prime_UTR	2903	3268	.	+	.	
ID=LOC_Os01g01010.1.MSUv7.0.five_prime_UTR.1;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	exon	3354	3616	.	+	.	ID=LOC_Os01g01010.1.MSUv7.0.exon.2;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	five_prime_UTR	3354	3448	.	+	.	
ID=LOC_Os01g01010.1.MSUv7.0.five_prime_UTR.2;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	CDS	3449	3616	.	+	0	ID=LOC_Os01g01010.1.MSUv7.0.CDS.1;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	exon	4357	4455	.	+	.	ID=LOC_Os01g01010.1.MSUv7.0.exon.3;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	CDS	4357	4455	.	+	0	ID=LOC_Os01g01010.1.MSUv7.0.CDS.2;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	exon	5457	5560	.	+	.	ID=LOC_Os01g01010.1.MSUv7.0.exon.4;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	CDS	5457	5560	.	+	0	ID=LOC_Os01g01010.1.MSUv7.0.CDS.3;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	exon	7136	7944	.	+	.	ID=LOC_Os01g01010.1.MSUv7.0.exon.5;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	CDS	7136	7944	.	+	1	ID=LOC_Os01g01010.1.MSUv7.0.CDS.4;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	exon	8028	8150	.	+	.	ID=LOC_Os01g01010.1.MSUv7.0.exon.6;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	CDS	8028	8150	.	+	2	ID=LOC_Os01g01010.1.MSUv7.0.CDS.5;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	exon	8232	8320	.	+	.	ID=LOC_Os01g01010.1.MSUv7.0.exon.7;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	CDS	8232	8320	.	+	2	ID=LOC_Os01g01010.1.MSUv7.0.CDS.6;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	exon	8408	8608	.	+	.	ID=LOC_Os01g01010.1.MSUv7.0.exon.8;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	CDS	8408	8608	.	+	0	ID=LOC_Os01g01010.1.MSUv7.0.CDS.7;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
Chr1	phytozomev11	exon	9210	9617	.	+	.	ID=LOC_Os01g01010.1.MSUv7.0.exon.9;Parent=LOC_Os01g01010.1.MSUv7.0;pacid=33123311
```

- **Objectif: Représenter graphiquement (avec Circos):**

1- Les chromosomes

2- La densité en éléments transposables

3- La densité en gènes

4- Données Bonus (libre)

5- La synténie intragénomique

4- Données Bonus (libre)

…à partir du fichier d’annotation

…à partir d’autres recherches de synténie

…autre

- **Objectif: Représenter graphiquement (avec Circos):**

1- Les chromosomes

2- La densité en éléments transposables

3- La densité en gènes

4- Données Bonus (libre)

5- La synténie intragénomique

5- La synténie intragénomique

Éditer un fichier texte tabulé <links>.txt

Chr1	2000000	4000000	Chr2	1500000	3000000
Chr1	5000000	5200000	Chr2	4000000	3700000
Chr1	11500000	12500000	Chr2	13500000	14000000
Chr2	3000000	3500000	Chr5	3000000	3500000
Chr3	4000000	4200000	Chr4	1000000	1200000
Chr3	4200001	5000000	Chr4	12000000	10000000
Chr3	6000000	8000000	Chr1	500000	1000000
Chr3	9000000	9100000	Chr1	1500000	1200000
Chr4	500	200000	Chr2	14500000	14600000
Chr5	11000000	11500000	Chr2	8000000	8500000

ID1 Début fin ID2 Début fin

5- La synténie intragénomique

Éditer un fichier texte tabulé <links>.txt

Chr1	2000000	4000000	Chr2	1500000	3000000
Chr1	5000000	5200000	Chr2	4000000	3700000
Chr1	11500000	12500000	Chr2	13500000	14000000
Chr2	3000000	3500000	Chr5	3000000	3500000
Chr3	4000000	4200000	Chr4	1000000	1200000
Chr3	4200001	5000000	Chr4	12000000	10000000
Chr3	6000000	8000000	Chr1	500000	1000000
Chr3	9000000	9100000	Chr1	1500000	1200000
Chr4	500	200000	Chr2	14500000	14600000
Chr5	11000000	11500000	Chr2	8000000	8500000

ID1 Début fin ID2 Début fin

…à l'aide des données de synténie intra-génomique (*.gcoords.txt)

Détection de blocs synténiques dupliqués (SynMap/DAGChainer)

Download Results

General	Homolog search	Diagonals	Results
Analysis Log	Fasta file for Sorghum bicolor (v3.1): CDS	DAGChainer Initial Input file	DAGChainer Output
Image File	Fasta file for Sorghum bicolor (v3.1): CDS	DAGChainer Input file converted to gene order	DAGChainer output in genomic coordinates
	Unfiltered Last results	DAGChainer Input file post repetitve matches filtered	Final syntenic gene-set output with GEvo links
	Filtered Last results (no tandem duplicates)		Condensed syntelog file with GEvo links
	Tandem Duplicates for Sorghum bicolor (v3.1)		
	Tandem Duplicates for Sorghum bicolor (v3.1)		
			SVG Version of Syntenic Dotplot

Send to SynMap3D: https://genomevolution.org/coge/SynMap3D.pl?x_gid=31607;y_gid=31607

Regenerate this analysis: https://genomevolution.org/r/oqtm

```
#
#1      480.0   a31607_1        b31607_1        f       10
a31607_1        1||48286845||48289028||Sobic.001G263500.1.v3.1||-1||CDS||985422562||3817||77.77 48289028        48286845        b31607_1
1||71614120||71622661||Sobic.001G437801.1.v3.1||1||CDS||985429282||6389||77.77  71614120        71622661        1.000000e-250   50
a31607_1        1||49414202||49415809||Sobic.001G264500.1.v3.1||1||CDS||985422597||3830||83.59  49414202        49415809        b31607_1
1||71631369||71633018||Sobic.001G438100.1.v3.1||1||CDS||985429290||6392||83.59  71631369        71633018        1.000000e-250   97
a31607_1        1||49722474||49725087||Sobic.001G264800.1.v3.1||1||CDS||985422608||3834||76.99  49722474        49725087        b31607_1
1||71639933||71640907||Sobic.001G438200.1.v3.1||1||CDS||985429293||6393||76.99  71639933        71640907        1.200000e-77    147
a31607_1        1||50296283||50298001||Sobic.001G265900.1.v3.1||-1||CDS||985422645||3847||74.3   50298001        50296283        b31607_1
1||71667654||71669060||Sobic.001G438400.1.v3.1||1||CDS||985429298||6395||74.3   71667654        71669060        4.300000e-78    194
a31607_1        1||50783046||50785022||Sobic.001G266400.1.v3.1||1||CDS||985422660||3852||74.19  50783046        50785022        b31607_1
1||71687096||71689051||Sobic.001G438800.1.v3.1||1||CDS||985429310||6399||74.19  71687096        71689051        7.300000e-193   244
a31607_1        1||50888462||50892684||Sobic.001G267100.1.v3.1||-1||CDS||985422685||3862||88.31 50892684        50888462        b31607_1
1||71698521||71703749||Sobic.001G439000.1.v3.1||-1||CDS||985429316||6401||88.31 71703749        71698521        1.000000e-250   294
a31607_1        1||51092331||51094282||Sobic.001G267800.1.v3.1||1||CDS||985422712||3872||76.53  51092331        51094282        b31607_1
1||71716400||71718713||Sobic.001G439100.1.v3.1||-1||CDS||985429322||6403||76.53 71718713        71716400        1.300000e-178   344
a31607_1        1||51321034||51325071||Sobic.001G268400.1.v3.1||1||CDS||985422735||3880||70.77  51321034        51325071        b31607_1
1||71761250||71765619||Sobic.001G439600.1.v3.1||1||CDS||985429347||6413||70.77  71761250        71765619        1.900000e-148   394
a31607_1        1||51754381||51754965||Sobic.001G269900.1.v3.1||1||CDS||985422779||3897||81.33  51754381        51754965        b31607_1
1||71793126||71793872||Sobic.001G439900.1.v3.1||1||CDS||985429361||6418||81.33  71793126        71793872        4.600000e-40    430
a31607_1        1||52223500||52228681||Sobic.001G270800.1.v3.1||1||CDS||985422802||3906||77.15  52223500        52228681        b31607_1
1||71797446||71801583||Sobic.001G440000.1.v3.1||-1||CDS||985429364||6419||77.15 71801583        71797446        9.400000e-237   480
#2      463.0   a31607_1        b31607_1        f       10
a31607_1        1||60785621||60787903||Sobic.001G320000.1.v3.1||1||CDS||985424756||4656||80.67  60785621        60787903        b31607_1
1||79365970||79368237||Sobic.001G529700.1.v3.1||1||CDS||985432835||7730||80.67  79365970        79368237        1.000000e-250   50
a31607_1        1||60795235||60801229||Sobic.001G320200.1.v3.1||1||CDS||985424762||4658||75.2   60795235        60801229        b31607_1
1||79375598||79379711||Sobic.001G529800.1.v3.1||1||CDS||985432838||7731||75.2   79375598        79379711        1.000000e-250   100
a31607_1        1||60801893||60803856||Sobic.001G320300.1.v3.1||-1||CDS||985424765||4659||80.33 60803856        60801893        b31607_1
1||79400314||79402539||Sobic.001G530300.1.v3.1||-1||CDS||985432853||7736||80.33 79402539        79400314        1.000000e-250   150
a31607_1        1||60805296||60805679||Sobic.001G320400.1.v3.1||-1||CDS||985424768||4660||77.27 60805679        60805296        b31607_1
1||79405777||79406127||Sobic.001G530350.1.v3.1||-1||CDS||985432856||7737||77.27 79406127        79405777        9.000000e-58    200
a31607_1        1||60810784||60814447||Sobic.001G320500.1.v3.1||-1||CDS||985424771||4661||66.67 60814447        60810784        b31607_1
1||79466234||79471748||Sobic.001G531000.1.v3.1||-1||CDS||985432882||7746||66.67 79471748        79466234        2.600000e-14    213
a31607_1        1||60885522||60889564||Sobic.001G321200.2.v3.1||-1||CDS||985424794||4670||78.06 60889564        60885522        b31607_1
1||79504966||79510277||Sobic.001G531700.1.v3.1||-1||CDS||985432907||7755||78.06 79510277        79504966        1.000000e-250   263
a31607_1        1||60904755||60906701||Sobic.001G321500.1.v3.1||-1||CDS||985424806||4675||77.14 60906701        60904755        b31607_1
1||79520401||79522981||Sobic.001G532000.1.v3.1||-1||CDS||985432915||7758||77.14 79522981        79520401        1.200000e-199   313
a31607_1        1||60925659||60927441||Sobic.001G321800.1.v3.1||-1||CDS||985424814||4678||74.59 60927441        60925659        b31607_1
1||79537584||79539995||Sobic.001G532200.1.v3.1||-1||CDS||985432921||7760||74.59 79539995        79537584        9.400000e-194   363
a31607_1        1||60947889||60949672||Sobic.001G322100.1.v3.1||-1||CDS||985424823||4681||81.0  60949672        60947889        b31607_1
1||79545417||79551114||Sobic.001G532300.1.v3.1||-1||CDS||985432924||7761||81.0  79551114        79545417        1.000000e-250   413
a31607_1        1||60960689||60970798||Sobic.001G322300.1.v3.1||-1||CDS||985424829||4683||71.39 60970798        60960689        b31607_1
1||79557423||79562902||Sobic.001G532400.1.v3.1||-1||CDS||985432929||7763||71.39 79562902        79557423        8.800000e-151   463
```

Détection de blocs synténiques dupliqués (SynMap/DAGChainer)

- **DAGChainer output**
  - Output generated by DAGChainer.

```
#1      133556.0        a4241_1 b7117_1 f       2673
a4241_1 1||1397989||1398285||ECDH10B_1366||-1||CDS||24909233||1293||99.7        1293    1293    b7117_1 1||1290089||1290385||E4401_1219||-1|
a4241_1 1||1398509||1399228||ECDH10B_1367||1||CDS||24909270||1294||100.0        1294    1294    b7117_1 1||1290594||1291328||E4401_1220||1||
```

- - DAGChainer output files contain type types of data lines:
    - "\#": lines separate identified syntenic blocks. The information contained in this line is:
      - 1. <syntenic block number> (NOTE: this is not a unique number: each chromosome-chromosome (contig-contig) comparison has this number reset)
      - Score for syntenic block
      - Organism A chromosome in syntenic block
      - Organism B chromosome in syntenic block
      - Orientation of syntenic block: f == same orientation, positive slope on dotplot; r == opposite orientation, negative slope on dotplot
      - Number of syntenic gene pairs identified in syntenic block
    - All other lines: contain a syntenic match
      - Org A Chromosome
      - Gene/region in org A
      - Start position in org A (may be relative gene order or absolute nucleotide position)
      - Stop position in org A (may be relative gene order or absolute nucleotide position)
      - Org B Chromosome
      - Gene/region in org B
      - Start position in org B (may be relative gene order or absolute nucleotide position)
      - Stop position in org B (may be relative gene order or absolute nucleotide position)
      - Evalue of match
      - Accumulative score of diagonal including this gene match (number increases with additional syntenic gene pairs within a syntenic block)
- **Merged DAGChainer output**
  - This file is generated if the analysis option is selected for merging neighboring syntenic regions
- **Quota Alignment output**
  - This file is generated if the analysis option for Quota Align is selected for enforcing a specific syntenic depth relationship between the analyzed genomes
- **DAGChainer output in genomic coordinates**
  - This file is the same as the DAGChainer output file but the genomic coordinates have been converted from gene order to nucleotide positions.

Editer le fichier de configuration <*circos.conf*>

../scripts/circos.conf

```
<<include etc/colors_fonts_patterns.conf>>
<<include ./ideogram.conf>>
<<include ./ticks.conf>>

<image>
angle_offset* = -82
<<include etc/image.conf>>
</image>
karyotype = ./karyotype.txt
chromosomes_units           = 1000000
chromosomes_display_default = yes


<plots>
<plot>
file       = ./plot_1.txt
max_gap    = 1u
color      = grey
min        = 1
max        = 12
r0         = 0.95r
r1         = 0.92r
thickness  = 2
fill_color = dblue
type       = histogram
</plot>

<plot>
file       = ./plot_2.txt
color   = spectral-7-div
scale_log_base = 0.6
max_gap = 1u
min     = 0
max     = 100000
r0         = 0.84r
r1         = 0.81r
type       = heatmap
</plot>
</plots>

<links>
<link>

file          = ./links.txt
color         = red
thickness     = 5
ribbon           = yes
flat             = yes
z      = 0
radius = 0.63r
bezier_radius = 0.4r
</link>
</links>

# Debugging, I/O an dother system parameters
# Included from Circos distribution.
<<include etc/housekeeping.conf>>
```

Appel de fichiers de config

Fichier « karyotype »

Plot_1

Plot_2

Plots

Link

Links

Run Circos

../scripts/circos_bash.sh

```
#! /bin/bash
#$ -S /bin/bash
#$ -M *.*@*univ-rennes1.fr
#$ -m bea
#$ -cwd
```

```
. /local/env/envperl-5.26.2.sh
. /local/env/envpython-3.9.5.sh
. /local/env/envcircos-0.69.8.sh
```

→ Sourcer les environnements

circos -conf circos.conf

Run Python sur Genouest (SLURM)

python_bash.sh

```
#! /bin/bash
#$ -S /bin/bash
#$ -M *.*@*univ-rennes1.fr
#$ -m bea
#$ -cwd

. /local/env/envpython-3.9.5.sh

python *.py fichier_a_traiter.*
```

- **CC2 (Rapport + scripts/fichiers)**

Rapport page 1

- (noms/prénoms du binôme)
- Image Circos + légendes

Rapport page 2

- Présentation des données et des méthodes

À rendre pour le : **06 avril 2026, 23:00**

le format SAM (mapping de lectures)

CIGAR

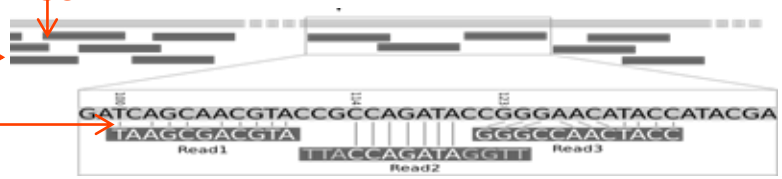
« 2M1D3M »

```
AAGTCTAGAA (ref)
  GT TAG (query)
```

POS

RNAME

QNAME



Version compressée : format BAM

MAPQ

Mesure la probabilité que la lecture soit mal positionnée

```
@HD VN:1.5 SO:coordinate
@SQ SN:ref LN:45
r001   99 ref  7 30 8M2I4M1D3M = 37  39 TTAGATAAAGGATACTG *
r002    0 ref  9 30 3S6M1P1I4M *  0   0 AAAAGATAAGGATA    *
r003    0 ref  9 30 5S6M       *  0   0 GCCTAAGCTAA       * SA:Z:ref,29,-,6H5M,17,0;
r004    0 ref 16 30 6M14N5M    *  0   0 ATAGCTTCAGC       *
r003 2064 ref 29 17 6H5M       *  0   0 TAGGC             * SA:Z:ref,9,+,5S6M,30,1;
r001  147 ref 37 30 9M         =  7 -39 CAGCGGCAT         * NM:i:1
```

Header section

Alignment section

Optional fields in the format of TAG:TYPE:VALUE

QUAL: read quality; * meaning such information is not available

SEQ: read sequence

TLEN: the number of bases covered by the reads from the same fragment. Plus/minus means the current read is the leftmost/rightmost read. E.g. compare first and last lines.

PNEXT: Position of the primary alignment of the NEXT read in the template. Set as 0 when the information is unavailable. It corresponds to POS column.

RNEXT: reference sequence name of the primary alignment of the NEXT read. For paired-end sequencing, NEXT read is the paired read, corresponding to the RNAME column.

CIGAR: summary of alignment, e.g. insertion, deletion

MAPQ: mapping quality

POS: 1-based position

RNAME: reference sequence name, e.g. chromosome/transcript id

FLAG: indicates alignment information about the read, e.g. paired, aligned, etc.

QNAME: query template name, aka. read ID